

Emmanuel Alabi

Software Engineer & AI/Robotics Researcher

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Research Interests

Embodied AI; robot learning; computer vision; autonomous systems; multimodal intelligence. Specifically, how an agent's representation of a physical environment should be shaped by having to move through it, and when learned components genuinely outperform well-specified classical methods.

Broader interests: machine learning, robotics, state estimation, planning, reinforcement learning.

Education

Babcock University — BSc (Hons) Software Engineering 2019 — 2023

Second Class Upper Division, CGPA 4.00/5.00. Ilishan-Remo, Ogun State, Nigeria. School of Computing and Engineering Sciences.

Relevant coursework: Algorithms and Data Structures; Discrete Mathematics; Introductory Statistics; Introduction to Operations Research; Artificial Intelligence and Applications; Introduction to Big Data Engineering; Computer Organization and Assembly Language; Operating System I; Database System Design, Implementation and Management.

Research

Risk-Calibrated Semantic Navigation Under Distribution Shift 2026

Preregistered ROS 2 / Gazebo / Nav2 benchmark. doi.org/10.5281/zenodo.22181941

- Calibration improved and clean-route efficiency held, but the preregistered collision-reduction and severity hypotheses did not: better component calibration did not establish safer navigation.
- Four systems differing only in the treatment component; protocol, manifest, thresholds and analysis code hash-locked before execution. 6,840 episodes; independent clean-clone reproduction byte-identical.
- emmanuelalabi.com/research/risk-calibrated-semantic-navigation

Driver Drowsiness Detection — Reproduction & Evaluation Study 2026

Independent computer vision study. github.com/alabiemmanuel177/driver-drowsiness-reproduction

- An oracle-classifier decomposition showed the historical temporal threshold, not residual classification error, dominated false alerts once classifier accuracy was sufficient.
- Subject-disjoint reproduction on the MRL Eye Dataset; CNN trained from scratch against a HOG+SVM baseline, with deterministic splits, automated leakage checks and three-seed ablations.
- emmanuelalabi.com/projects/driver-drowsiness-reproduction

Research Training

Independent Research Programme — Embodied & Multimodal Autonomy 2026 — present

- Self-directed programme in the mathematical, machine-learning and robotics foundations for independent research: linear algebra, probability, optimisation, state estimation and rigid-body geometry, each implemented from first principles.
- Reproducing published baselines in perception and navigation before extending them.
- emmanuelalabi.com/research/research-programme-2026

Selected Academic Projects

BUCODEL — Learning Management System for Open & Distance e-Learning 2022 — 2023

Undergraduate Final-Year Project (SENG490, 72/100), Babcock University. Supervisor: Dr. Stephen Maitanmi.

- Co-designed and implemented a web-based learning management system for Babcock University's Centre for Open and Distance e-Learning, supporting student, lecturer and administrator workflows.
- Built course and resource management, assignments, authentication, user administration, collaborative course activity and role-specific interfaces using React, Node.js/Express and MongoDB.
- Conducted requirements analysis, system design, iterative implementation under a spiral methodology, and interface/usability, unit and integration testing.
- emmanuelalabi.com/projects/bucodel-learning-management-system

Selected Technical Experience

Founder & CEO — Obelo

Oct 2025 — present

- Production AI platform: multi-agent orchestration over a structured brand representation, retrieval-augmented context, model routing across model families, asynchronous orchestration, and an automated brand-consistency grader checking every output before release.
- emmanuelalabi.com/projects/obelo-multimodal-ai-infrastructure

Software Engineer — Punch Group

Jan 2025 — Jul 2025

- Shipped three production applications: GraphQL optimisation and memory-leak resolution on a Next.js/Express product, a lead-generation tool with HubSpot and Salesforce integrations, and a real-time support service with in-app video.

Software Engineer — SPay Business

Nov 2023 — Dec 2024

- Production mobile-financial systems: authentication improvements, application performance, interface redesign, CI/CD automation and over-the-air mobile deployments.

Software Engineer — Union Bank of Nigeria

Jul 2023 — Nov 2023

- Enterprise applications in React and Java; designed and enforced website security protocols; delivered merchant notification systems over REST APIs.

Technical Skills

Research computing Python, PyTorch, NumPy, ROS 2, Gazebo, Nav2, statistical evaluation, uncertainty calibration, experiment design

AI systems Computer vision, multimodal systems, retrieval-augmented generation, LLM and model integration, evaluation systems

Software & infrastructure TypeScript, SQL, Docker, CI/CD, PostgreSQL, MongoDB, Node.js, React, Next.js

References

References available on request.